The Metric Bundle Programme: Research Roadmap Toward a Geometric Theory of Everything

From the Observerse to Emergent Spacetime
via Torsion, Free Energy, and Inference

Research Programme Document

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Abstract

This document lays out a structured research programme aimed at developing the metric bundle framework into a complete geometric theory unifying gravity, gauge interactions, matter, and observation. The programme is organised into four phases spanning approximately 5–10 years, with concrete milestones, go/no-go decision points, and specific calculations at each stage. It synthesises results from two companion papers—one on the gauge structure and spinorial content of the metric bundle, and one on the identification of torsion with variational free energy—and identifies the critical path from current results to testable predictions. The immediate priority is the Kaluza–Klein reduction of the 14-dimensional metric bundle to 4-dimensional effective physics, which serves as the primary validation test for the entire programme.

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1 Executive Summary: The State of Play

1.1 What Exists

Two mathematical papers have established:

Paper 1: Gauge Structure of the Metric Bundle

R1 The metric bundle $Y = \mathrm{Met}(X)$ over a 4-manifold X is a natural 14-dimensional space.

R2 The Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ arises as the intersection of the maximal compact reduction $K = \mathrm{Spin}(6) \times \mathrm{Spin}(4)$ and the complex structure reduction $\mathrm{U}(5)$ within $\mathrm{SO}(6,4)$.

R3 The normal bundle spinor $S(N)$ after $\mathrm{Spin}(6) \times \mathrm{Spin}(4)$ reduction is $16_{\mathbb{C}}$-dimensional, matching one generation of Standard Model fermions.

R4 The Pati–Salam group $\mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ is anomaly-free.

R5 An explicit action $S = S_{\mathrm{grav}} + S_{\mathrm{gauge}} + S_{\mathrm{Dirac}} + S_{\mathrm{tor}}$ is written down.

R6 The augmented torsion $T = \pi - \varepsilon^{-1}(d_{A_0}\varepsilon)$ transforms covariantly under the tilted gauge group.

Paper 2: Torsion as Variational Free Energy

R7 The augmented torsion admits a decomposition into prediction error + model update, formally identical to variational free energy.

R8 The gradient flow of S_{tor} with respect to the shift field is variational Bayesian inference.

R9 This connects to Sakthivadivel's gauge-theoretic Bayesian mechanics.

R10 The quantum FEP of Fields–Friston–Glazebrook–Levin provides the spacetime-free substrate.

R11 Under the inferential interpretation, the quantisation problem is dissolved: the Einstein–Hilbert action is an effective inferential functional, not a fundamental Lagrangian.

R12 Five concrete, testable conjectures are stated.

1.2 What's Missing

Critical Gaps

G1 Kaluza–Klein reduction: The action on Y^{14} has not been reduced to effective 4D physics. (BLOCKING — nothing else matters until this is done.)

G2 Gauge coupling prediction: The metric bundle geometry should fix coupling ratios at unification. Not computed.

G3 Higgs mechanism: Complex structure moduli on N as Higgs candidate. Not verified.

G4 Three generations: Only one generation emerges naturally. No mechanism for three.

G5 Emergence derivation: The claim that spacetime emerges from inferential dynamics is philosophical, not mathematical.

G6 Quantum corrections: No calculation of loop effects in the inferential framework.

G7 Cosmological constant: The "residual torsion" conjecture is untested.

G8 Novel prediction: No prediction that differs from the Standard Model + GR.

2 Phase 1: Validation (Months 1–12)

Goal: Determine whether the metric bundle framework produces correct 4D physics.

Go/No-Go: If the Kaluza–Klein reduction fails to produce Einstein + Pati–Salam + torsion with correct signs, the programme is terminated.

2.1 Task 1.1: Kaluza–Klein Reduction (Priority: CRITICAL)

The Calculation

Input: The 14-dimensional metric bundle Y with chimeric metric G_{AB}, a metric section $g : X \to Y$, and the full action $S[G, A, \Psi]$ on Y.
Output: The effective 4-dimensional action $S_{\text{eff}}[g_{\mu\nu}, A^a_\mu, \psi, \phi]$ obtained by integrating over the 10-dimensional fibre.
Method: Standard Kaluza–Klein ansatz, adapted to the metric bundle geometry.

2.1.1 Step 1: Coordinate System on Y

Let x^μ ($\mu = 0, 1, 2, 3$) be coordinates on X and y^m ($m = 1, \ldots, 10$) be coordinates on the fibre $\mathcal{F}_x = \text{Met}_x(X) \cong \text{GL}(4, \mathbb{R})/\text{O}(4)$.

A convenient parameterisation of the fibre: at each x, a metric $g_{\mu\nu}(x)$ is a positive-definite symmetric matrix, which we can write as

$$g_{\mu\nu}(x) = (\bar{g} + h)_{\mu\nu}(x), \tag{1}$$

where $\bar{g}_{\mu\nu}$ is the background (the section about which we expand) and $h_{\mu\nu}$ are the 10 fibre coordinates. But this mixes base and fibre; we need a more intrinsic description.

The fibre at x is $\mathcal{F}_x \cong P_x/K_x$ where $P_x = \mathrm{GL}(T_xX) \cong \mathrm{GL}(4,\mathbb{R})$ is the frame bundle fibre and $K_x = \mathrm{O}(T_xX, \bar{g}_x) \cong \mathrm{O}(4)$ is the isotropy group. The tangent space to the fibre is

$$T_{[e]}\mathcal{F}_x \cong \mathfrak{p}/\mathfrak{k} \cong \mathfrak{gl}(4)/\mathfrak{o}(4) \cong S^2(\mathbb{R}^4), \tag{2}$$

the space of 4×4 symmetric matrices. We choose coordinates y^m $(m = 1, \ldots, 10)$ corresponding to the 10 independent components of a symmetric matrix.

2.1.2 Step 2: The Chimeric Metric in Block Form

The chimeric metric on Y has the standard Kaluza–Klein block form:

$$ds_Y^2 = G_{AB}\, dz^A\, dz^B = g_{\mu\nu}\, dx^\mu\, dx^\nu + g_{mn}(A)\,(dy^m + A^m_\mu\, dx^\mu)(dy^n + A^n_\nu\, dx^\nu), \tag{3}$$

where:

- $g_{\mu\nu}(x)$ is the spacetime metric (from the section);
- g_{mn} is the Frobenius metric on the fibre (the DeWitt metric restricted to one fibre);
- $A^m_\mu(x)$ is the Kaluza–Klein gauge field, valued in the Lie algebra of the isometry group of the fibre.

The Frobenius (DeWitt) metric on the fibre is:

$$g_{mn}\, dy^m\, dy^n = \left(\bar{g}^{\mu\rho}\bar{g}^{\nu\sigma} - \frac{1}{2}\bar{g}^{\mu\nu}\bar{g}^{\rho\sigma}\right) dg_{\mu\nu}\, dg_{\rho\sigma}. \tag{4}$$

Key question: What is the isometry group of $(\mathcal{F}, g_{\mathrm{DeWitt}})$?

The DeWitt metric on $\mathrm{GL}(4)/\mathrm{O}(4)$ is a left-invariant metric. The isometry group acts by $g \mapsto A\, g\, A^T$ for $A \in \mathrm{GL}(4)$, giving isometry group $\mathrm{GL}(4,\mathbb{R})$. The maximal compact subgroup is $\mathrm{O}(4)$, acting as $g \mapsto O\, g\, O^T$.

For the metric bundle, the relevant isometry group after imposing orientation and positivity is $\mathrm{SO}^+(4)$, which under complexification and restriction gives:

$$\mathrm{SO}(4) \cong [\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R]/\mathbb{Z}_2. \tag{5}$$

But the full structure group we found in Paper 1 is $\mathrm{Spin}(6) \times \mathrm{Spin}(4)$ from the Ricci decomposition splitting the normal bundle. This arises not from the isometry of a single fibre but from the structure of the *normal bundle* to the section.

2.1.3 Step 3: The Scalar Curvature of Y

The central calculation. For a Kaluza–Klein metric of the form (3), the scalar curvature of Y decomposes as [**?**]:

$$\boxed{R_Y = R_X + R_{\mathcal{F}} - \frac{1}{4}F^m_{\mu\nu}F^{m\,\mu\nu} - \frac{1}{2}(\nabla_\mu\phi)(\nabla^\mu\phi) + \cdots} \tag{6}$$

where:

- R_X is the scalar curvature of the base (spacetime);
- $R_{\mathcal{F}}$ is the scalar curvature of the fibre;
- $F^m_{\mu\nu} = \partial_\mu A^m_\nu - \partial_\nu A^m_\mu + f^m_{np} A^n_\mu A^p_\nu$ is the Yang–Mills field strength;
- ϕ represents scalar moduli (the "breathing mode" of the fibre).

The Einstein–Hilbert action on Y therefore gives:

$$\frac{1}{16\pi G_{14}} \int_Y R_Y \, \mathrm{vol}_Y = \frac{V_{\mathcal{F}}}{16\pi G_{14}} \int_X \left[R_X - \frac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu} + R_{\mathcal{F}} + \cdots \right] \mathrm{vol}_X, \tag{7}$$

where $V_{\mathcal{F}} = \int_{\mathcal{F}} \mathrm{vol}_{\mathcal{F}}$ is the fibre volume.

This is the payoff: from pure 14-dimensional geometry, we get Einstein gravity + Yang–Mills gauge theory on X, with the gauge group determined by the isometry group of the fibre.

2.1.4 Step 4: Identifying the Gauge Group

This is the crucial step where the metric bundle may differ from a generic KK compactification. The fibre $\mathcal{F} = \mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$ is a 10-dimensional non-compact symmetric space. Its isometry group is *not* compact, so the naive KK gauge group is non-compact.

However, Paper 1 showed that the relevant structure is not the full isometry group but the structure group of the *normal bundle* N to the metric section, which reduces to $\mathrm{Spin}(6) \times \mathrm{Spin}(4)$ and then to $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

Problem 1 (Critical computation)**.** *Compute the spectrum of the Laplacian $\Delta_{\mathcal{F}}$ on $\mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$ with DeWitt metric. The harmonic analysis on this symmetric space determines which fields appear in the KK tower and their masses. The zero modes are the massless fields in 4D.*

Specifically, determine:

(a) The zero modes of $\Delta_{\mathcal{F}}$ acting on scalar functions: these give 4D scalar fields.

(b) The Killing vectors of $(\mathcal{F}, g_{DeWitt})$: these determine the gauge group.

(c) The zero modes of the Dirac operator on $\mathcal{F}$ (if a spin structure exists): these give 4D fermions.

(d) The quadratic Casimir of the gauge group in the adjoint representation: this determines the gauge coupling normalisation.

2.1.5 Step 5: Concrete Implementation Plan

1. **Week 1–2**: Set up the symmetric space $\mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$ in Mathematica/SageMath. Compute the Cartan decomposition $\mathfrak{gl}(4) = \mathfrak{o}(4) \oplus \mathfrak{p}$ where $\mathfrak{p} \cong S^2(\mathbb{R}^4)$.

2. **Week 3–4**: Compute the DeWitt metric on $\mathfrak{p}$ explicitly. The metric is $\langle h, k \rangle = \mathrm{tr}(h \cdot k) - \frac{1}{2}\,\mathrm{tr}(h)\,\mathrm{tr}(k)$ on symmetric matrices. Compute Christoffel symbols.

3. **Week 5–8**: Compute the Riemann tensor and scalar curvature $R_{\mathcal{F}}$ of the fibre. This is known for symmetric spaces: $R_{\mathcal{F}} = -\frac{1}{4} \sum_{m,n} |[e_m, e_n]_{\mathfrak{p}}|^2$ where $\{e_m\}$ is an orthonormal basis of $\mathfrak{p}$ and $[\cdot,\cdot]_{\mathfrak{p}}$ is the $\mathfrak{p}$-component of the bracket.

4. **Week 9–12**: Compute the Killing vectors of $(\mathcal{F}, g_{\text{DeWitt}})$. These are the infinitesimal isometries $V^m(y)$ satisfying $\nabla_{(m}V_{n)} = 0$. The Killing algebra is the isometry Lie algebra.

5. **Week 13–16**: Assemble the KK reduction (7). Identify the 4D fields, gauge group, and coupling constants. Compare to Standard Model + GR.

6. **Week 17–20**: Include the torsion sector. Show that S_{tor} on Y reduces to a specific 4D torsion action.

2.2 Task 1.2: Fibre Geometry (Priority: HIGH)

The symmetric space $\mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$ is well-studied in the mathematics literature. Key facts to verify or compute:

Property	**Expected**	**Status**
Dimension	10	Known
Rank	4	Known
Curvature sign	Non-positive	Known (non-compact type)
Isometry group	$\mathrm{GL}(4,\mathbb{R})$	Known
Maximal compact subgroup	$\mathrm{SO}(4) \cong [\mathrm{SU}(2) \times \mathrm{SU}(2)]/\mathbb{Z}_2$	Known
Volume	Infinite (non-compact)	Known — requires regularisation
Spin structure	Exists?	Unknown — must check
Dirac spectrum	?	Must compute
Scalar Laplacian spectrum	Continuous	Known (non-compact $\Rightarrow$ no gap)

The Non-Compactness Problem

The fibre $\mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$ is non-compact, so:

(1) The fibre volume $V_{\mathcal{F}}$ is infinite. The KK reduction (7) requires regularisation.

(2) The Laplacian has continuous spectrum. There is no mass gap between zero modes and massive KK modes.

(3) The Killing vectors generate a non-compact gauge group.

Possible resolutions:

(a) The physical fibre is not $\mathrm{GL}^+(4)/\,\mathrm{SO}(4)$ but a compact quotient or submanifold thereof. E.g., metrics with bounded curvature, or metrics in a fixed conformal class.

(b) The "compactification" is dynamical: the potential from $R_{\mathcal{F}}$ localises fluctuations near the section, creating an effective mass gap.

(c) The inferential interpretation resolves this: the "fibre volume" is the normalisation of the prior, which is finite for a proper generative model.

This is the single biggest technical obstacle in the KK reduction.

2.3 Task 1.3: Gauge Coupling Prediction (Priority: HIGH)

If the KK reduction succeeds, the gauge coupling constants at the compactification scale are determined by:

$$\frac{1}{g_a^2} = \frac{V_a}{16\pi G_{14}}, \tag{8}$$

where V_a is the volume of the orbit of the gauge group G_a in the fibre. For the Pati–Salam group $\mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$:

$$\frac{g_4^2}{g_L^2} = \frac{V_L}{V_4}, \quad \frac{g_4^2}{g_R^2} = \frac{V_R}{V_4}. \tag{9}$$

The ratio g_L/g_R tests left-right symmetry. If $V_L = V_R$ (the fibre metric respects parity), then $g_L = g_R$ at unification, which is the standard prediction of left-right symmetric models.

The ratio g_4/g_L determines the $\mathrm{SU}(3) \times \mathrm{U}(1)$ coupling ratio at the Pati–Salam scale, which can be compared to the observed running of α_s and α_{em}.

Problem 2 (Gauge coupling computation)**.** *Compute the volumes V_4, V_L, V_R of the* $\mathrm{SU}(4)$, $\mathrm{SU}(2)_L$, $\mathrm{SU}(2)_R$ *orbits in* $\mathrm{GL}^+(4,\mathbb{R})/\mathrm{SO}(4)$ *with respect to the DeWitt metric.*

2.4 Decision Point 1 (Month 12)

GO/NO-GO Decision

Proceed if:

- KK reduction produces R_X+ Yang–Mills + fermions + torsion with correct signs.
- The gauge group contains Pati–Salam (or SM directly).
- Gauge coupling ratios are compatible with observed running (within a factor of 2).

Terminate if:

- Wrong signs in the gauge kinetic term (ghosts).
- Gauge group does not contain SM.
- Coupling ratios are off by an order of magnitude or more.

Redirect if:

- Non-compactness problem is insurmountable with current fibre $\Rightarrow$ investigate compact quotients.
- Right structure but wrong details $\Rightarrow$ modify the fibre metric or add flux.

3 Phase 2: Mechanism (Months 12–36)

Goal: Establish the three key mechanisms: symmetry breaking, generation replication, and the torsion–inference connection.

3.1 Task 2.1: Higgs Mechanism from Complex Structure Moduli

The normal bundle N to the metric section admits complex structures $J : N \to N$ with $J^2 = -1$. The moduli space $\mathcal{M}_J$ of such complex structures is the space of "Higgs-like" scalar fields.

Problem 3. *(a) Compute the dimension of $\mathcal{M}_J$. Does it match the SM Higgs sector (one complex doublet = 4 real components)?*

(b) Compute the effective potential $V(J)$ induced by the KK reduction. Does it have a minimum that breaks Pati–Salam $\to$ SM?

(c) Compute the mass of fluctuations around the minimum. Does it match the observed Higgs mass (125 GeV) for reasonable parameter values?

3.2 Task 2.2: Three Generations

Four candidate mechanisms:

(a) Index theory: The index of the Dirac operator on Y restricted to the section. If $\mathrm{ind}(D_Y|_g) = 3$, the three generations are topological.

(b) Division algebras: The Furey programme gives $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, with $\mathrm{Cl}_6(\mathbb{C})$ giving one generation. Three copies might arise from the three associative subalgebras of $\mathbb{O}$, or from $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H}$ acting on $\mathbb{O}$.

(c) Multi-scale Markov blankets: In the inferential interpretation, three generations = three levels of the inference hierarchy. This is speculative but testable: compute the number of independent eigenmodes of $\Delta_{\mathcal{F}}$ in the first non-trivial band.

(d) Topological invariant of X: The third Stiefel–Whitney class $w_3(X)$ or some other characteristic class of X equals 3 for physically relevant 4-manifolds.

Problem 4. *Investigate mechanism (a) first: compute ind$(D_Y|_g)$ for the metric bundle over S^4, T^4, and $\mathbb{R}^4$ using the Atiyah–Singer index theorem.*

3.3 Task 2.3: Torsion–Free Energy Correspondence (Rigorous)

Paper 2 established the correspondence at the level of formal structure. Phase 2 should make it rigorous:

Problem 5. *(a) Define a measure μ on $\mathrm{Met}(X)$ such that the torsion action equals the KL divergence: $S_{tor} = [q\|p]$ exactly (not just formally).*

(b) Show that the Euler–Lagrange equations of the full action $S = S_{grav} + S_{gauge} + S_{Dirac} + S_{tor}$ are equivalent to the FEP equations of motion for a specific generative model.

(c) Compute the Fisher information metric on the parameter space of the action and show it equals the DeWitt metric (or a rescaling thereof).

3.4 Decision Point 2 (Month 36)

GO/NO-GO Decision

Proceed if: Higgs mechanism works, at least one generation mechanism is viable, torsion–FE correspondence is rigorous.
Redirect if: Higgs doesn't work $\Rightarrow$ investigate alternative symmetry breaking. Generation mechanism unclear $\Rightarrow$ focus on the two that are most tractable.

4 Phase 3: Prediction (Months 36–72)

Goal: Extract novel, testable predictions that distinguish this framework from the Standard Model + GR.

4.1 Task 3.1: Pati–Salam Phenomenology

If the framework predicts Pati–Salam rather than SU(5) GUT:

- **Proton decay**: No proton decay at the SU(5) rate ($\tau_p \gg 10^{36}$ yr). Currently consistent with experiment.
- **B−L violation**: Possible at higher energies. Look for n-$\bar{n}$ oscillation.
- **Right-handed W_R**: Predict the mass of the right-handed W boson from the Pati–Salam breaking scale, which is determined by the complex structure moduli potential.
- **Leptoquarks**: Pati–Salam leptoquarks at the PS scale.

4.2 Task 3.2: Torsion Signatures

- **Gravitational wave dispersion**: Torsion introduces a frequency-dependent phase velocity for gravitational waves: $v(f) = 1 + \alpha_T (f/f_P)^2 + \cdots$. The coefficient α_T is determined by the torsion coupling λ.
- **Baryogenesis**: The torsion spin-spin contact interaction in the early universe provides a CP-violating mechanism for baryogenesis, potentially explaining the observed baryon asymmetry.
- **Cosmological constant**: If Conjecture 8 from Paper 2 holds, $\Lambda = \langle T_{\text{res}}, T_{\text{res}} \rangle / \lambda$. Computing this requires the topology of the actual metric bundle of our universe.

4.3 Task 3.3: Gauge Coupling Unification

The most decisive prediction: the metric bundle geometry determines the gauge coupling ratios at unification. Run the SM couplings up to the Pati–Salam scale and compare with the geometric prediction from Task 1.3.

If the prediction is correct to within $\sim 1\%$, this would be extraordinarily strong evidence.

4.4 Decision Point 3 (Month 72)

PUBLISH/PIVOT Decision

Publish major results if: At least one novel prediction is made and is compatible with existing data. Seek experimental tests.
Pivot if: All predictions are either trivially satisfied or contradicted. The framework reproduces known physics but adds nothing new.

5 Phase 4: Foundations (Months 72–120)

Goal: Establish the rigorous foundations for emergent spacetime and the consciousness connection.

5.1 Task 4.1: The Emergence Derivation

This is the hardest and most important long-term goal:

Problem 6 (The Emergence Problem)**.** *Starting from:*

(i) A pre-geometric substrate (ruliad, tensor network, or abstract computational structure);

(ii) Markov blankets defined on that substrate;

(iii) Free energy minimisation as the dynamical principle;

derive:

(iv) An emergent smooth 4-manifold X;

(v) An emergent metric bundle $Y = \mathrm{Met}(X)$;

(vi) The effective action S_{eff} on Y matching the result of Phase 1.

This is the mathematical equivalent of deriving thermodynamics from statistical mechanics—showing that the macroscopic equations (GR + SM) are the statistical limit of the microscopic dynamics (FEP on the ruliad).

5.2 Task 4.2: The Quantum Sector

Even if the UV problem is "dissolved," there are still quantum effects that need to be reproduced:

- The path integral over metrics ⇒ becomes a sum over inference strategies?
- Loop corrections to gauge couplings ⇒ higher-order corrections to the variational approximation?
- The conformal anomaly ⇒ related to the information-geometric curvature of the statistical manifold?

5.3 Task 4.3: The Consciousness Interface

The weakest link: connecting the mathematical framework to conscious experience. This requires:

- Making Hoffman's conscious agent theory mathematically precise enough to embed in the metric bundle framework.
- Showing that the "perceptual interface" of a conscious agent is literally a metric section $g : X \to Y$.
- Deriving the specific dimensionality (4D spacetime, 10D fibre) from properties of the conscious agent network.

This may be the work of a generation, not a single research programme.

6 Summary: The Critical Path

Priority	Task	Timeline	Difficulty
1	KK reduction of Y^{14}	3–6 months	Hard
2	Gauge coupling ratios	6–12 months	Medium
3	Complex structure moduli / Higgs	12–24 months	Hard
4	Three generations	12–36 months	Very hard
5	Torsion–FE rigorous proof	12–24 months	Medium
6	Torsion observational signatures	24–48 months	Medium
7	Emergence derivation	48–120 months	Extremely hard
8	Novel testable prediction	When ready	—

The critical path runs through Tasks $1 \to 2 \to 3/5 \to 8$. If Task 1 fails, nothing else matters. If it succeeds, Tasks 2 and 3 become the priority, with Task 5 providing the theoretical framework.

The single most important result that could come out of this programme is a *correct, specific prediction for the gauge coupling ratios at unification.* This would be the equivalent of Mercury's perihelion for general relativity—a single number, derived from pure geometry, that matches observation.

7 Appendix: The KK Reduction — Detailed Setup

We now begin the actual computation. This appendix lays out the mathematical framework; the computation itself will be continued in a dedicated technical note.

7.1 The Symmetric Space $\mathrm{GL}^+(4,\mathbb{R})/\mathrm{SO}(4)$

7.1.1 Cartan Decomposition

The Lie algebra $\mathfrak{gl}(4,\mathbb{R})$ decomposes under the Cartan involution $\theta(X) = -X^T$ as:

$$\mathfrak{gl}(4,\mathbb{R}) = \underbrace{\mathfrak{o}(4)}_{\mathfrak{k}} \oplus \underbrace{S^2(\mathbb{R}^4)}_{\mathfrak{p}}, \tag{10}$$

where $\mathfrak{k} = \{X \in \mathfrak{gl}(4) : X = -X^T\}$ (antisymmetric matrices) and $\mathfrak{p} = \{X \in \mathfrak{gl}(4) : X = X^T\}$ (symmetric matrices).

A basis for $\mathfrak{p}$: let E_{ij} be the elementary matrix with 1 in position (i, j) and 0 elsewhere. Then

$$e_{(ij)} = \begin{cases} E_{ii} & \text{if } i = j, \\ \frac{1}{\sqrt{2}}(E_{ij} + E_{ji}) & \text{if } i < j, \end{cases} \tag{11}$$

gives 10 basis elements: $e_{(11)}, e_{(12)}, e_{(13)}, e_{(14)}, e_{(22)}, e_{(23)}, e_{(24)}, e_{(33)}, e_{(34)}, e_{(44)}$.

7.1.2 The DeWitt Metric

The DeWitt supermetric on $\mathfrak{p}$ is:

$$G(h, k) = \mathrm{tr}(h \cdot k) - \frac{1}{2}\,\mathrm{tr}(h)\ \mathrm{tr}(k) \tag{12}$$

for $h, k \in \mathfrak{p} \cong S^2(\mathbb{R}^4)$.

In the basis $\{e_{(ij)}\}$, the metric components are:

$$G_{(ij),(kl)} = G(e_{(ij)}, e_{(kl)}) = \mathrm{tr}(e_{(ij)} \cdot e_{(kl)}) - \frac{1}{2}\,\mathrm{tr}(e_{(ij)})\ \mathrm{tr}(e_{(kl)}). \tag{13}$$

Computing explicitly:

$$G_{(ii),(jj)} = \delta_{ij} - \frac{1}{2} = \begin{cases} \frac{1}{2} & i = j \\ -\frac{1}{2} & i \neq j \end{cases}, \tag{14}$$

$$G_{(ij),(ij)} = 1 \quad (i < j), \tag{15}$$

$$G_{(ij),(kl)} = 0 \quad \text{if } \{i, j\} \neq \{k, l\}. \tag{16}$$

The diagonal-diagonal block is a 4×4 matrix with $\frac{1}{2}$ on the diagonal and $-\frac{1}{2}$ off-diagonal:

$$G_{\text{diag}} = \frac{1}{2}\begin{pmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{pmatrix}. \tag{17}$$

This has eigenvalues 2 (eigenvector $(1, -1, -1, -1, \ldots)$ etc.) and $-\frac{1}{2} \cdot 3 + \frac{1}{2} = -1$ (eigenvector $(1, 1, 1, 1)$). Wait—let me recompute.

The eigenvalues of G_{diag}: the matrix $M = \frac{1}{2}(I_4 - J_4)$ where J_4 is the all-ones matrix. The eigenvalues of J_4 are 4 (eigenvector $(1, 1, 1, 1)$) and 0 (multiplicity 3). So:

$$\text{eigenvalues of } G_{\text{diag}} = \frac{1}{2}(1 - 4) = -\frac{3}{2} \quad (\times 1), \quad \frac{1}{2}(1 - 0) = \frac{1}{2} \quad (\times 3). \tag{18}$$

The DeWitt metric has indefinite signature on the diagonal sector. The trace mode $\mathrm{tr}(h) = h_{11} + h_{22} + h_{33} + h_{44}$ has negative norm. This is the conformal factor. The DeWitt metric on $\mathrm{Met}(X)$ has signature $(9, 1)$ at each point—it is Lorentzian on the space of metrics, with the conformal mode being the "timelike" direction.

This is actually a known and important feature: the conformal factor problem in quantum gravity. The Euclidean path integral over metrics $\int \mathcal{D}g\, e^{-S_{\text{EH}}}$ is ill-defined because the conformal factor has wrong-sign kinetic term.

The Conformal Factor Problem

The DeWitt metric on $\mathrm{GL}^+(4)/\mathrm{SO}(4)$ has signature $(9,1)$, not $(10,0)$. The "time-like" direction is the conformal factor $\Omega = (\det g)^{1/4}$.

In the inferential interpretation: The conformal mode is special because it determines the *volume* of the experienced spacetime. Rescaling the metric $g \to \Omega^2 g$ changes the overall scale of the agent's perceptual interface but not its local geometry. The negative norm of this mode reflects the fact that the "cost" of the conformal factor is negative—larger universes are "less surprising" than smaller ones, which is thermodynamically natural (more states available in larger volumes).

Possible resolution: The path integral should be performed with a contour rotation on the conformal mode, or the conformal factor should be treated as a gauge degree of freedom (as in unimodular gravity, where $\det g = 1$ is imposed).

7.1.3 Killing Vectors

The Killing vectors of $(\mathrm{GL}^+(4)/\mathrm{SO}(4), G_{\mathrm{DeWitt}})$ generate the isometry algebra. For a symmetric space G/K, the isometry algebra is $\mathfrak{g}$ itself. Here $\mathfrak{g} = \mathfrak{gl}(4,\mathbb{R})$, which is 16-dimensional.

The Killing vectors split under $\mathfrak{k} \oplus \mathfrak{p}$:

- $\mathfrak{k} = \mathfrak{o}(4)$: 6 Killing vectors generating rotations $g \mapsto OgO^T$.
- $\mathfrak{p} = S^2(\mathbb{R}^4)$: 10 Killing vectors generating translations in the fibre.

The 6-dimensional compact part $\mathfrak{o}(4)$ decomposes as $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$. This gives gauge fields:

$$A^a_\mu \in \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R, \quad a = 1, \ldots, 6. \tag{19}$$

But we need $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$**, which is 12-dimensional.** The isometry group of a single fibre only gives $\mathrm{SO}(4) \cong [\mathrm{SU}(2) \times \mathrm{SU}(2)]/\mathbb{Z}_2$, which is too small.

Resolution: The full gauge group does not come from the isometry of a single fibre. It comes from the structure group of the *bundle*—the group of fibre-preserving diffeomorphisms of Y. For a homogeneous bundle $Y = G \times_K \mathcal{F}$, this is the full structure group G. In our case, the relevant structure group is determined by the normal bundle decomposition, yielding $\mathrm{Spin}(6) \times \mathrm{Spin}(4)$ as established in Paper 1.

This means the KK reduction is not of the standard type (isometry of internal space = gauge group). Instead, it's closer to a reduction of the *frame bundle*, where the structure group is the full orthogonal group of the fibre tangent space.

Modified KK Ansatz

The correct KK reduction for the metric bundle uses:

$$\text{Gauge group} = \text{Structure group of } N \to X = \text{Spin}(6) \times \text{Spin}(4), \tag{20}$$

not the isometry group of a single fibre. The gauge connection A^a_μ is the connection on the normal bundle, not a Kaluza–Klein gauge field from off-diagonal metric components.
The scalar curvature then decomposes as:

$$R_Y = R_X + R_N - \frac{1}{4}|F_N|^2 + (\text{scalar moduli}) + (\text{torsion}), \tag{21}$$

where F_N is the curvature of the normal bundle connection and R_N is the intrinsic curvature of the normal bundle.
This is a *submanifold reduction* rather than a *compactification reduction*: the base is a submanifold of the total space (via the section $g : X \to Y$), not a factor in a product.

This insight—that the reduction is submanifold-type rather than product-type—is crucial and may be the key innovation needed to make the KK reduction work despite the non-compactness of the fibre.

7.2 Next Steps for the Computation

The immediate next computation (to be continued in a dedicated technical note) is:

1. **The Gauss equation**: For the section $g : X \hookrightarrow Y$, the Gauss equation relates R_Y, R_X, the second fundamental form II, and the normal curvature $R^\perp_N$:

$$R_Y = R_X + 2\,\text{Ric}_Y(\nu, \nu) + |H|^2 - |\text{II}|^2 + R^\perp_N, \tag{22}$$

   where H is the mean curvature and ν denotes normal directions. **This is the correct starting point**, not the standard KK decomposition.

2. **The second fundamental form**: Compute II for the metric section $g : X \to Y$. This measures how the section curves within Y, and its norm squared is closely related to the torsion action.

3. **The normal curvature**: Compute $R^\perp_N$ for the normal bundle N. This gives the Yang–Mills term.

4. **Assembly**: Put it all together to get the effective 4D action.